

HTTP Solved Problems

Problem #1

A webpage contains 6 objects, each of size 100 KB.

Given:

- TCP RTT = 80 ms
- HTTP RTT = 40 ms
- Bandwidth = 10 Mbps

Calculate:

1. Total load time using non-persistent HTTP
2. Total transmission time only (ignore RTT)
3. Percentage of time spent in connection overhead

Solution:

For non-persistent connection

$$\begin{aligned} \text{Load time} &= \text{Total RTT} + \frac{\text{Total data}}{BW} \\ &= \text{No. of objects}(\text{TCP RTT} + \text{HTTP RTT}) + \frac{\text{No. of objects} * \text{size of each objects}}{BW} \\ &= 6(80 + 40) + \frac{6 * 100 * 10^3 * 8}{10 * 10^6} * 1000 = 720 + 480 = 1200 \text{ ms} \end{aligned}$$

[transmission time has been multiplied by 1000 to represent it in ms as the Total RTT is in ms]

$$\text{Total transmission delay} = \frac{\text{No. of objects} * \text{size of each objects}}{BW}$$

$$= \frac{6 * 100 * 10^3 * 8}{10 * 10^6} * 1000 = 480 \text{ ms}$$

Percentage of time spent in connection overhead = Total RTT * 100 / load time

$$= 720 * 100 / 1200 = 60\%$$

Problem #2

A webpage contains 10 objects, each of size 250 KB.

Given:

- TCP RTT = 90 ms
- Bandwidth = 25 Mbps
- Total measured load time (non-persistent) = 3.2 seconds

Find:

1. The value of HTTP RTT.
2. The percentage of total time spent in data transmission.
3. If HTTP RTT is reduced by 50%, compute the new load time.

Solution:

1. For non-persistent connection,

$$\text{Load time} = \text{No. of objects}(\text{TCP RTT} + \text{HTTP RTT}) + \frac{\text{No. of objects} * \text{size of each objects}}{BW}$$

$$3200 = 10(90 + \text{HTTP RTT}) + \frac{10 * 250 * 10^3 * 8}{25 * 10^6} * 1000$$

$$\text{HTTP RTT} = 150 \text{ ms}$$

2. transmission time = $\frac{10 * 250 * 10^3 * 8}{25 * 10^6} * 1000 = 800 \text{ ms}$

$$\% \text{ of total time spent in transmission} = \frac{800}{3200} * 100 = 25\%$$

3. HTTP RTT=150 ms.

50% of the HTTP RTT=75 ms

$$\text{New load time} = 10(90 + 75) + \frac{10 * 250 * 10^3 * 8}{25 * 10^6} * 1000 = 1650 \text{ ms}$$

Problem #3

A browser needs to load an HTML page that includes references to the following resources using a **non-persistent HTTP** connection (*The network link has a bandwidth of T*):

- Base HTML file, taking **321 ms** to download
- 3 embedded videos, each taking **250 ms** to download
- 5 Java applets, each taking **111 ms** to download

Each of the three control packets, such as **SYN packet**, **SYN-ACK packet**, and **ACK-HTTP request packet**, incurs a one-way delay of **64 ms**. In contrast, an HTTP Response experiences a one-way delay of **48 ms**. All transfers occur at a constant rate, with no use of parallel TCP connections or HTTP pipelining.

- Calculate the **total round-trip time** for loading the HTML Page.
- Compute the **total time required** to fetch the HTML page
- State, mathematically, how the total time in (II) changes if the **link bandwidth T is halved**.

Solution:

TCP RTT = SYN_Delay + SYN_HTTP Req Delay

$$= 64 + 64 = 128 \text{ ms}$$

HTTP RTT = ACK_HTTP Req Delay + HTTP Response Delay

$$= 64 + 48 = 112 \text{ ms}$$

Total no. of objects = 1 + 3 + 5 = 9

- For non-persistent HTTP connection

Total RTT = No. of objects * (TCP RTT + HTTP RTT)

$$= 9(128 + 112) = 2160 \text{ ms}$$

- Total transmission time = $1 * 321 + 3 * 250 + 5 * 111 = 1626 \text{ ms}$ [Also known as object transmission time (OTT)]

Load time or total time to fetch the page = Total RTT + Total Transmission Time

$$= 2160 + 1626 = 3786 \text{ ms}$$

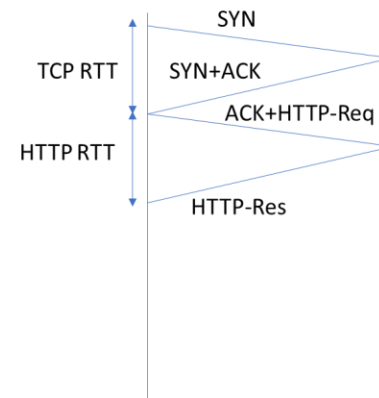
- Given bandwidth is T .

New bandwidth is half of its previous. So the new bandwidth = $T/2$

Load time = Total RTT + Total transmission time

$$= \text{Total RTT} + \frac{\text{Total Data}}{BW}$$

$$= \text{Total RTT} + \frac{\text{Total Data}}{T/2}$$



$$= Total\ RTT + \frac{2 * Total\ Data}{T}$$

That is the total transmission time will be twice the previous total transmission time, while the total RTT will not change.

Problem #4

Asif loads an HTML page in Google Chrome over a **persistent HTTP connection** (TCP termination time and the third phase of handshaking are negligible). The page contains **11 objects** in total, **including** the base HTML file. The size of each object is **4 MB**. The **total round-trip time** to load the page is **455 ms**. Note that the **TCP RTT** is **twice** the **per-object HTTP request-and-response RTT**. The server's bandwidth is **16 Mbps**.

- I. Calculate the **TCP RTT** and the **per-object HTTP request response RTT**.
- II. Calculate the **total page load time**.
- III. If the page had 20 objects instead of 11, explain which part of the delay would increase the most in persistent HTTP, and why.

Solution:

I.

Total RTT = TCP RTT + No. of objects * HTTP RTT

455 = 2 * HTTP RTT + 11 * HTTP RTT = 13 HTTP RTT

HTTP RTT = 455 / 13 = 35 ms

TCP RTT = 2 * HTTP RTT = 2 * 35 = 70 ms

- II.
$$Load\ time = TCP\ RTT + No.\ of\ objects * HTTP\ RTT + \frac{No.\ of\ objects * size\ of\ each\ objects}{BW}$$

$$Load\ time = 70 + 11 * 35 + \frac{11 * 4 * 10^6 * 8}{16 * 10^6} * 1000 = 22,455\ ms$$

- III. Per object transmission delay = $\frac{4 * 10^6 * 8}{16 * 10^6} * 1000 = 2000\ ms$

The HTTP RTT is 35 ms while the per-object transmission delay is 2000 ms.

Thus, if the number of objects increases from 11 to 20, the transmission time will increase much more compared to the total RTT.

Question# 5

A university LAN has a bandwidth of 100 Mbps. Students generate an average of 50 requests per second, with each requested object having an average size of 1.5 Mbits.

- Calculate the Traffic Intensity on the LAN.
- Based on your result, is the LAN currently in the "No congestion area" or the "Congestion area"?

Solutions:

$$\text{Traffic intensity or utilization} = \frac{\text{Request arrival rate} * \text{Object size}}{BW} = \frac{50 * 1.5}{100} = 0.75$$

Since traffic intensity or utilization < 1, the network is not congested.

Question# 6

A network administrator measures the following average delays for a webpage request:

- LAN Delay: 15 ms
- Access Delay: 450 ms
- Internet Delay: 2.2 seconds

Calculate the total Average Response Time in milliseconds.

Suppose the network installs a web cache.

- 40% of requests are "hits" and are served locally with a total delay of only 15 ms.
- 60% of requests are "misses" and experience the full delay of 2665 ms.
- Calculate the new weighted Average Response Time.

Solution:

Average delay without Proxy server

Delay = LAN Delay + Access Delay + Internet Delay = 15 + 450 + 2200 = 2665 ms

Average delay with Proxy server: $0.4 * 15 + 0.6 * 2665 = 1605$ ms

Question# 7

You are given the following data for a branch office:

- LAN Bandwidth: 1 Gbps (1000 Mbps)
- Access Link Bandwidth: 20 Mbps
- Traffic: 18 requests/sec of 1 Mbit each.

A) Which link is the "bottleneck" (the one closest to congestion)?

B) At what request rate (requests/sec) will the Access Link intensity reach exactly 1.0 (Capacity)?

Solution:

A)

$$\text{Traffic intensity of LAN} = \frac{\text{Arrival rate} * \text{Object size}}{BW} = \frac{18 * 1}{1000} = 0.018$$

$$\text{Traffic intensity of Access Link} = \frac{\text{Arrival rate} * \text{Object size}}{BW} = \frac{18 * 1}{20} = 0.9$$

Since traffic intensity of the access link is close to 1. This will limit this speed of the link. Hence, the access link is the bottleneck of the network.

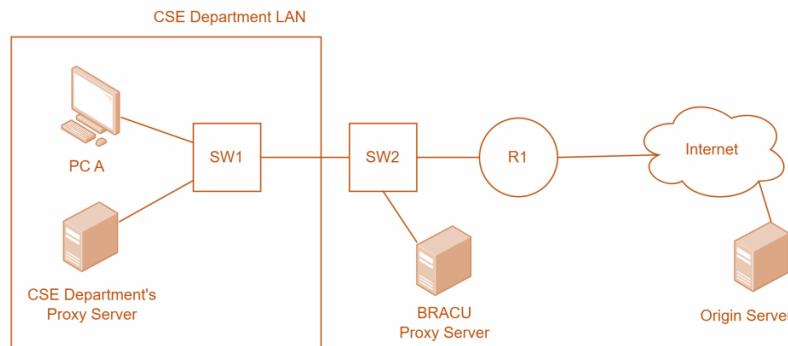
B)

$$\text{Traffic intensity of Access Link} = \frac{\text{request rate} * \text{Object size}}{BW}$$
$$1 = \frac{\text{Request rate} * 1}{20}$$

Request rate=20

Question# 7

Each department of Brac University has dedicated proxy servers. If content is not found in the department's proxy server, it is searched into Brac University's central proxy server and then to the origin server. The department's proxy server can handle **40% of** the requests, and the remaining **60%** of the requests are **equally divided and resolved** by the BRACU Proxy and the Origin servers. Given that the CSE Department network's LAN



delay is **15 ms**, the BRACU network's LAN delay is **30 ms**, the access delay for the origin server is **100 ms**, and the internet delay is **200 ms**.

- I. Calculate the average response time for a webpage.
- II. A device from the **EEE** department of Brac University, has just visited the webpage xyz.com. Identify the **exact response time** for PCA if it wants to visit the webpage

Solution:

Delay for LAN proxy: 15 ms

Delay for the Univesity proxy: 15+30 =45 ms

Delay for the origin server=15+30+100+200=345 ms

Average response time=0.4*15+0.3*45+0.3*345=123 ms

Exact response time for PCA to visit xyz.com=45 ms

Question# 8

A user at Brac University is accessing a research paper from an origin server. The department has a proxy server that implements web caching to reduce Access Link intensity.

At 10:00 AM, the proxy server fetches a document with the following header: Last-Modified: Wed, 25 Feb 2026 09:00:00 GMT. The cache is configured to consider items "stale" after 2 hours. At 1:00 PM, a student requests the same document.

Questions:

1. **Request Type:** Describe the specific type of HTTP request the proxy server will send to the origin server to verify if the cached copy is still valid.
2. **Response Handling:** If the document has not changed on the origin server, what specific HTTP response code will the origin server return, and will the "Internet Delay" (200 ms) or the "LAN Delay" (15 ms) be the dominant factor in the student's perceived response time? Justify your answer using the delay values provided in the BracU diagram.
3. **Intensity Impact:** If the hit ratio of the proxy increases because many students are accessing the same stale-but-unchanged document, how does this affect the Traffic Intensity of the Access Link?.

Solution:

1. Conditional GET
2. 304 NOT modified. Between LAN and Access delay, LAN delay will be dominant as the object will be delivered from the proxy, and the transmission of the tiny message (conditional GET) introduces much lower delay (experienced by the access link).
3. The access link traffic intensity will reduce.

NOTE: Kindly solve as many relevant previous semester questions.
as possible.